Implicit Differentiation and Inverse Functions

ANSWER KEY

Section 1: First Derivative by Implicit Differentiation

1 Find $\frac{dy}{dx}$ for the circle $x^2 + y^2 = 25$.

a. $-\frac{x}{y}$ ✓

b. $\frac{x}{y}$

c. $-\frac{y}{x}$

d. $\frac{y}{x}$

Differentiate implicitly: $2x + 2y\frac{dy}{dx} = 0 \Rightarrow \frac{dy}{dx} = -\frac{x}{y}$.

2 Find $\frac{dy}{dx}$ for the curve $x^3 + y^3 = 6xy$.

a. $\frac{2y - x^2}{y^2 - 2x}$ ✓

b. $\frac{x^2 - 2y}{2x - y^2}$

c. $\frac{2x - y^2}{x^2 - 2y}$

d. $\frac{x^2 + y^2}{6}$

$3x^2 + 3y^2y' = 6(y + xy') \Rightarrow 3y^2y' - 6xy' = 6y - 3x^2 \Rightarrow y'(y^2 - 2x) = 2y - x^2 \Rightarrow y' = \frac{2y - x^2}{y^2 - 2x}$.

3 Find the slope of the tangent line to $x^2y + y^2 = 5$ at the point $(2, 1)$.

a. $-\frac{2}{3}$ ✓

b. $-\frac{4}{5}$

c. $-\frac{4}{6}$

d. -1

$2xy + x^2y' + 2yy' = 0$. At $(2, 1)$: $2(2)(1) + 4y' + 2(1)y' = 0 \Rightarrow 4 + 6y' = 0 \Rightarrow y' = -\frac{4}{6} = -\frac{2}{3}$.

Section 2: Second Derivatives Implicitly

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4 For $x^2 + y^2 = 16$, what is $\frac{d^2y}{dx^2}$ expressed solely in terms of y?

a. $-\frac{16}{y^3}$ ✓

b. $\frac{16}{y^3}$

c. $-\frac{1}{y^2}$

d. $-\frac{4}{y^3}$

$y' = -\frac{x}{y}$. Differentiate: $y'' = -\frac{1(y) - xy'}{y^2} = -\frac{y - x(-x/y)}{y^2} = -\frac{y^2 + x^2}{y^3}$. Since $x^2 + y^2 = 16$, $y'' = -\frac{16}{y^3}$.

5 At the point $(0, 2)$ on the ellipse $4x^2 + y^2 = 4$, what is $\frac{d^2y}{dx^2}$?

a. -2 ✓

b. 2

c. -4

d. 0

$8x + 2yy' = 0 \Rightarrow y' = -\frac{4x}{y}$. At $(0, 2)$, $y' = 0$. $y'' = -\frac{4y - 4xy'}{y^2} = -\frac{4(2) - 0}{4} = -\frac{8}{4} = -2$.

6 What is the sign of the concavity at $(3, 4)$ on the circle $x^2 + y^2 = 25$?

a. **Negative (concave down)** ✓

b. Positive (concave up)

c. Zero (inflection point)

d. Undefined

$y'' = -\frac{25}{y^3}$. At $(3, 4)$, $y'' = -\frac{25}{64} < 0$, which is concave down.

Section 3: Derivatives of Inverse Functions

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7 If g is the inverse of f, which formula gives $g'(a)$ where $f(b) = a$?

a. $g'(a) = \frac{1}{f'(b)}$ ✓

b. $g'(a) = f'(b)$

c. $g'(a) = \frac{1}{f'(a)}$

d. $g'(a) = -\frac{1}{f'(b)}$

By the Inverse Function Theorem, $g'(a) = \frac{1}{f'(f^{-1}(a))} = \frac{1}{f'(b)}$.

8 Let $f(x) = x^3 + 2x - 1$. If g is the inverse of f, find $g'(2)$.

a. $\frac{1}{5}$ ✓

b. 5

c. $\frac{1}{14}$

d. 14

Find x where $f(x) = 2$: $x^3 + 2x - 1 = 2 \Rightarrow x^3 + 2x - 3 = 0 \Rightarrow x = 1$. Then $f'(x) = 3x^2 + 2 \Rightarrow f'(1) = 5$. Thus $g'(2) = \frac{1}{f'(1)} = \frac{1}{5}$.

9 Let $f(x) = x^5 + x + 1$. What is the derivative of f^{-1} at $x = 3$?

a. $\frac{1}{6}$ ✓

b. 6

c. $\frac{1}{80}$

d. $\frac{1}{5}$

Notice $f(1) = 1 + 1 + 1 = 3$. $f'(x) = 5x^4 + 1 \Rightarrow f'(1) = 6$. Therefore, $(f^{-1})'(3) = \frac{1}{6}$.

Section 4: Derivatives of Inverse Trigonometric Functions

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10 What is $\frac{d}{dx}[\arcsin(3x)]$?

a. $\frac{3}{\sqrt{1-9x^2}}$ ✓

b. $\frac{1}{\sqrt{1-9x^2}}$

c. $\frac{3}{1+9x^2}$

d. $\frac{3}{\sqrt{1-3x^2}}$

Formula: $\frac{d}{dx}[\arcsin u] = \frac{u'}{\sqrt{1-u^2}}$. Here $u = 3x$, $u' = 3$, giving $\frac{3}{\sqrt{1-9x^2}}$.

11 Find $\frac{d}{dx}[\arctan(x^2)]$.

a. $\frac{2x}{1+x^4}$ ✓

b. $\frac{1}{1+x^4}$

c. $\frac{2x}{1+x^2}$

d. $\frac{2x}{\sqrt{1-x^4}}$

Formula: $\frac{d}{dx}[\arctan u] = \frac{u'}{1+u^2}$. Here $u = x^2$, $u' = 2x$, giving $\frac{2x}{1+x^4}$.

12 What is the slope of the tangent line to $y = \arctan(x)$ at $x = 1$?

a. $\frac{1}{2}$ ✓

b. 1

c. $\frac{\pi}{4}$

d. $\frac{1}{\sqrt{2}}$

$y' = \frac{1}{1+x^2}$. At $x = 1$: $y'(1) = \frac{1}{1+1^2} = \frac{1}{2}$.

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